the stone hits the Earth, it warms it: its mechanical energy is transformed into heat. And there is no way back from there. If the second law of thermodynamics did not exist, if heat did not exist, if there existed no microscopic swarming, the stone would rebound perpetually; it would never land and be still.

It is entropy, not energy, that keeps stones on the ground and the world turning.

The entire coming into being of the cosmos is a gradual process of disordering, like the pack of cards that begins in order and then becomes disordered through shuffling. There are no immense hands that shuffle the universe. It does this mixing by itself, in the interactions among its parts that open and close during the course of the mixing, step by step. Vast regions remain trapped in configurations that remain ordered, until here and there new channels are opened through which disorder spreads.[99]

What causes events to happen in the world, what writes its history, is the irresistible mixing of all things, going from the few ordered configurations to the countless disordered ones. The entire universe is like a mountain that collapses in slow motion. Like a structure that very gradually crumbles.

From the most minute events to the more complex ones, it is this dance of ever-increasing entropy, nourished by the initial low entropy of the universe, that is the real dance of Shiva, the destroyer.

TRACES AND CAUSES

The fact that entropy has been low in the past leads to an important fact that is ubiquitous and crucial for the difference between past and future: the past leaves traces of itself in the present.

Traces are everywhere. The craters of the moon testify to impacts in the past. Fossils show the forms of living creatures from long ago. Telescopes show how far off galaxies were in the past. Books contain our history; our brains swarm with memories.

Traces of the past exist, and not traces of the future, *only* because entropy was low in the past. There can be no other reason, since the only source of the difference between past and future is the low entropy of the past.

In order to leave a trace, it is necessary for something to become arrested, to stop moving, and this can happen only in an irreversible process—that is to say, by degrading energy into heat. In this way, computers heat up, the brain heats up, the meteors that fall into the moon heat it; even the goose quill of a medieval scribe in a Benedictine abbey heats a little the page on which he